Peter Julius M. Estacio

Grouped with:

Augusto Gabbriel M. Calilung

Rafael Luis Mari R. Lambo

ENGG 27.01 - M

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Progress Report 3 for Project 2: Power Method for Eigenvalues and Eigenvectors

As of March 13, 2026, major progress has been made with Augusto Gabbriel M. Calilung and Rafael Luis Mari R. Lambo as groupmates. The program function for calculating the Eigenvalues and Eigenvector of an input matrix has been tested. Additionally, the program is now able to take user input, validate said input, and read a matrix file according to the project specifications. Specifically, the readMatrixFile() function is able to open a filestream based on a user input file name, and dynamically store the matrix values into a 2d vector initialized in the main program. Additionally, a console interface has been developed to accept user input and display the Power Method calculations. Further work then includes allowing the user to opt in or out of displaying the individual traces of the Power Method calculations, along with displaying the final results to the maximum precision available for a double. A screenshot of some of the program results is shown below. This screenshot has been cut for brevity, as the full program output is too long.

```
Hello, welcome to find the Eigenvalue and Eigenvector-inator!
Please input the name of a matrix file like 'testMatrix': testMatrix
Matrix file 'testMatrix' opened successfully.
Success! Detected a square 5x5 matrix.
==========================================
n = 0
==========================================
For l = 0, eigenvalue = 9.6
For l = 1, eigenvalue = 13.8
For l = 2, eigenvalue = -21.6795
For l = 3, eigenvalue = -13.3114
For l = 4, eigenvalue = -14.986
For l = 5, eigenvalue = -14.5694
For l = 6, eigenvalue = -14.3675
For l = 7, eigenvalue = -14.3038
For l = 8, eigenvalue = -14.4376
For l = 9, eigenvalue = -14.3054
For l = 10, eigenvalue = -14.3647
For l = 11, eigenvalue = -14.3477
For l = 12, eigenvalue = -14.3476
For l = 13, eigenvalue = -14.3475
For l = 14, eigenvalue = -14.3496
For l = 15, eigenvalue = -14.347
For l = 16, eigenvalue = -14.3486
For l = 17, eigenvalue = -14.348
For l = 18, eigenvalue = -14.3481
For l = 19, eigenvalue = -14.3481
For l = 20, eigenvalue = -14.3481
For l = 21, eigenvalue = -14.3481
For l = 22, eigenvalue = -14.3481
For l = 23, eigenvalue = -14.3481
For l = 24, eigenvalue = -14.3481
For l = 25, eigenvalue = -14.3481
For l = 26, eigenvalue = -14.3481
For l = 27, eigenvalue = -14.3481
For l = 28, eigenvalue = -14.3481
For l = 29, eigenvalue = -14.3481
For l = 30, eigenvalue = -14.3481
For l = 31, eigenvalue = -14.3481
For l = 32, eigenvalue = -14.3481
For l = 33, eigenvalue = -14.3481
For l = 34, eigenvalue = -14.3481
For l = 35, eigenvalue = -14.3481
For l = 36, eigenvalue = -14.3481
For l = 37, eigenvalue = -14.3481
For l = 38, eigenvalue = -14.3481
For l = 39, eigenvalue = -14.3481
For l = 40, eigenvalue = -14.3481
For l = 41, eigenvalue = -14.3481
For l = 42, eigenvalue = -14.3481
For l = 43, eigenvalue = -14.3481
For l = 44, eigenvalue = -14.3481
For l = 45, eigenvalue = -14.3481
For l = 46, eigenvalue = -14.3481
For l = 47, eigenvalue = -14.3481
For l = 48, eigenvalue = -14.3481
For l = 49, eigenvalue = -14.3481
For l = 50, eigenvalue = -14.3481
```

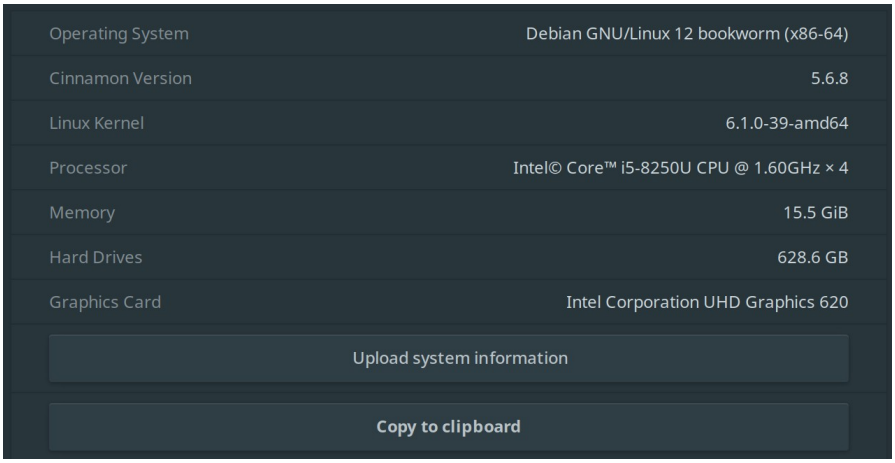
sample screenshots – program executable run and example results

```
* Executing task: C/C++: LINUX BUILD (g++)

Starting build...
/usr/bin/g++ -fdiagnostics-color=always -g *.cpp -o '/home/pedr0/Documents/Coding/ENGG 27 - 2/Project 2/main'

Build finished successfully.
* Terminal will be reused by tasks, press any key to close it.
```

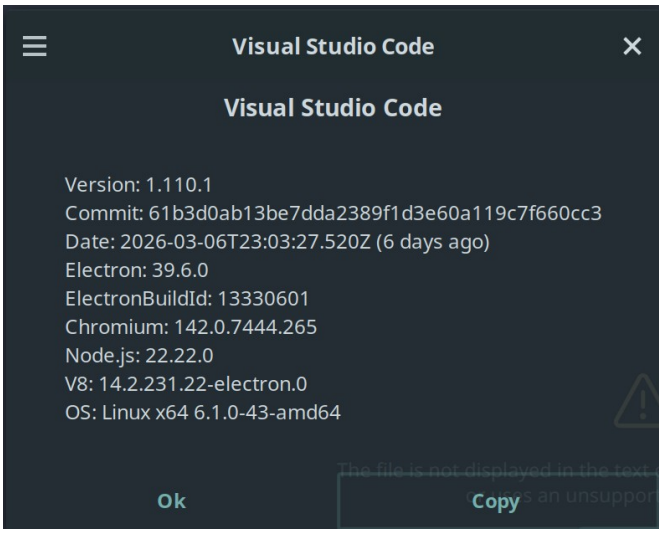
sample screenshot – compilation on Visual Studio Code

```
pedr0@pedr0:~/Documents/Coding$ g++ --version
g++ (Debian 12.2.0-14+deb12u1) 12.2.0
Copyright (C) 2022 Free Software Foundation, Inc.
This is free software; see the source for copying conditions.  There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
```

compiler version – g++ (Debian 12.2.0-14+deb12u1) 12.2.0



operating system – Debian 12 bookworm, Cinnamon version 5.6.8



IDE – Visual Studio Code ver 1.110.0